

A Novel 8-Transistor Full Adder with Reduced Power Consumption using 32nm CNTFET

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Abstract— Using cutting-edge 32 nm Carbon Nanotube Field-Effect Transistor (CNTFET) technology, a unique 8-transistor full adder is suggested for very large scale integration (VLSI) systems. This design attempts to solve the difficulties of scaling conventional complementary metal-oxide-semiconductor (CMOS) based adders into nanoscale technology while striking a balance between power consumption, physical size, and operational speed. In HSPICE simulations, the design exhibits a fast propagation delay of 5.55×10^{-11} seconds while maintaining performance across latency, power consumption, and energy efficiency with a low transistor count. This delay is constant over a broad supply voltage range (0.7 V to 1.5 V). In comparison to current designs like HFA26, CMOS28, and 9T adders, the average power consumption is still low, and the PDP and EDP values are good. Additionally, steady power and delay performance under realistic thermal circumstances is confirmed by temperature change studies. Monte Carlo analysis confirms that the design is resilient to changes in the process. The suggested 8T full adder is a promising option for next nanoscale and low-power VLSI applications since it provides a small, low-power, and energy-efficient solution.

Keywords— Energy-Delay Product (EDP), Power-Delay Product (PDP), Low Power VLSI, Full Adder, CNTFET, and Nanoscale Design.

I. INTRODUCTION

Digital circuit functions including division, subtraction, and multiplication are based on the basic mathematical process of addition [1]. Since a 1-bit complete adder is the fundamental component of binary adders, its efficient design is a main focus in contemporary Very-Large-Scale Integration (VLSI) applications where speed and efficiency are crucial [2, 3]. Minimizing power consumption, lowering latency, and improving overall performance are the main goals of the several complete adder designs that have been created over time employing different logic styles and technologies [4]. The primary method for improving the performance and packing density

of digital systems has been circuit shrinking. But there are serious drawbacks to scaling CMOS devices to nanoscale dimensions, including as short-channel effects, inadequate gate control, and high leakage power, all of which reduce circuit reliability and efficiency. Therefore, the low-power and high-speed demands of modern VLSI designs are difficult for standard CMOS logic to achieve [5]. Due to these constraints, researchers are now investigating alternate device technologies for digital circuit design, including Single-Electron Transistors (SET) [6], Multi-Gate FETs (MUGFET) [7], and Carbon Nanotube FETs (CNTFET) [8], Quantum Dot Cellular Automata (QCA) [9]. The necessity for energy-efficient adders has also increased due to the growing demand for battery-powered products [10]. However, physical limitations raise the likelihood of errors when device size decreases to the atomic level. This is consistent with Moore's rule, which states that as transistor density increases exponentially, manufacturing costs, design complexity, and leakage problems increase [11–13]. The efficiency of complete adders in CMOS technology has been improved by a variety of design approaches [13–17]. To cut down on power dissipation, methods such as Modified Gate Diffusion Input (MGDI), Dual-Value Logic (DVL), Transmission Gate Logic (TGL), Static CMOS, and Pass Transistor Logic (PTL) have been developed. These methods do, however, frequently have limitations, including higher latency, limited output swing, and increased area overhead [18]. In order to overcome these drawbacks, this work explores a mixed-logic approach that combines the benefits of GDI and other logic types to suggest devices with enhanced performance and fewer transistors. Three GDI-based adders are specifically examined: a new 8T design suggested in this work, a modified 9T design for enhanced output swing, and an existing 12T design for comparison. These designs undergo a thorough evaluation and comparison with traditional adders that are built using various logic techniques [19].

II. EXISTING DESIGN

2.1 Hybrid Full Adder

A 26-transistor hybrid full adder, the HFA26 has been regarded as a high-performance option for arithmetic circuits in the literature. Speed is given top priority in its design, and post-layout simulation experiments have demonstrated that HFA26 continuously performs better in terms of propagation delay than a large number of other low-power adders. When run under nominal supply conditions, it has actually been found to be up to 34% faster than more recent suggestions like PBFA26. Its decreased logic depth and effective signal propagation pathways are the primary causes of this speed advantage [14].

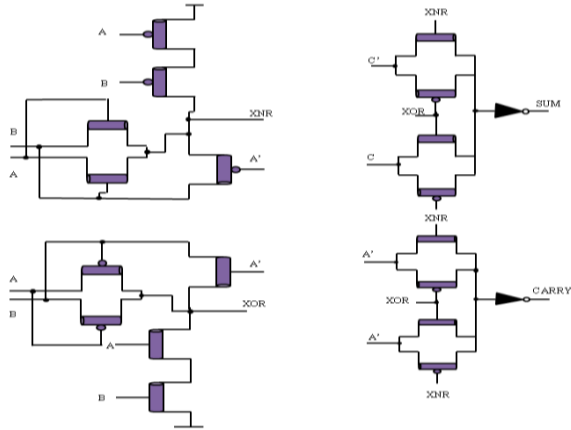


Fig. 1. Hybrid Full Adder. [14]

HFA26's main drawback, however, is its asymmetric signal route delays, which cause timing discrepancies between the creation of carry and total signals. Particularly during specific input transition scenarios, mismatches in the system cause glitches—brief, undesirable signal variations—at the overall output. When included into large-scale multiplier architectures or cascaded adders, these bugs lead to needless false switching activity, which not only raises dynamic power dissipation but also weakens the design's resilience. Although HFA26 is appropriate for high-speed applications, its tendency to have glitches makes it less appealing for low-power systems where energy efficiency is crucial.

2.2 Traditional Full Adder

In digital VLSI design, the traditional 28-transistor static CMOS full adder, which is represented by the CMOS28 design, has long been used as a reference standard. Because of its symmetric pull-up and pull-down networks, CMOS28 is one of the most reliable adders among the reference designs and is naturally resistant to errors. The literature's simulation findings show that CMOS28 has the lowest power consumption throughout a broad working voltage range, with claimed savings of around 10–14% when compared to PBFA26 at nominal supply levels and near-threshold. For energy-constrained systems where leakage and dynamic power are major issues, this makes CMOS28 very appealing.

However, there are two significant problems with the CMOS28 circuit. First, compared to other modern implementations like BFA22 and PBFA26, the design calls

for 28 transistors, which is more. The silicon area is increased by this comparatively larger device count, which might affect scalability in extremely small systems. Second, compared to high-speed circuits like HFA26, its delay performance is only moderate, while being reliable. Consequently, CMOS28 is not the best option for situations where operational speed is the key design issue, even if it offers an outstanding balance of power efficiency and reliability[10].

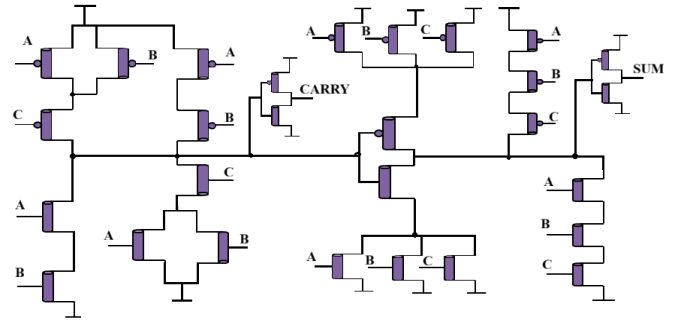


Fig. 2. Traditional CMOS28. [22]

2.3 Proposed Bridge Full Adder

The PBFA26 full adder's recently created 26-transistor implementation gets beyond some of the drawbacks of current circuits. The addition of an internal delay component (transistors M1 through M4) is a major innovation of this design. This component greatly lowers the possibility of spurious glitches at the final output—a recurring issue with the HFA26 design—by ensuring that the carry input (C) is precisely synchronized with the signals utilized for the XOR-based sum generation. Additionally, the sum-carry generating stage's intrinsic capacitances function as natural filters, efficiently absorbing these brief signal spikes and improving overall switching stability. [21]

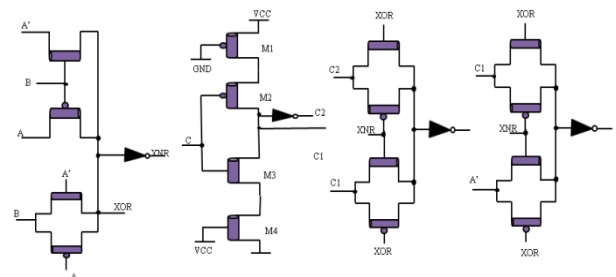


Fig. 3. Proposed Bridge Full Adder. [21]

A good balance between power consumption and operating speed is offered by the PBFA26 full adder. Although it uses less power than the BFA22, TFA22, and RFL22 designs, its energy efficiency is marginally lower than that of CMOS28 at a 1.2 Volt operation. When included into bigger systems, such cascaded multi-bit adders or Booth multiplier structures, the PBFA26 is the more energy-efficient option due to its greater glitch tolerance, even though the HFA26 is still quicker in individual, independent simulations. Indeed, PBFA26 exhibits lower power consumption than CMOS28 when included into 16-bit cascaded chains, underscoring its applicability for large-scale arithmetic circuits. All things considered, CMOS28 provides outstanding power efficiency and resilience at the cost of a

greater transistor count and considerable latency, whereas HFA26 is a high-speed but glitch-prone architecture. A more realistic trade-off between speed, power, and glitch immunity is offered by the suggested PBFA26 circuit, which makes use of design-level synchronization and decreased complexity. Because of its effectiveness in cascaded and multiplier topologies, PBFA26 is a strong and competitive contender for nanoscale CNTFET-based VLSI implementations, even though it does not perform better than CMOS28 or HFA26 in every standalone parameter.

2.4 GDI Based Full Adder

In order to achieve low device count, full swing operation, and decreased power-delay product, the 9-transistor full adder depicted above is developed utilizing a mix of Gate Diffusion Input (GDI), modified-GDI (mGDI), and Transmission Gate Logic (TGL). The main idea is to use a tiny TGL-based XOR/XNOR core to generate the propagate signal $X=A \oplus B$. This core produces both X and X bar simultaneously with minimal delay and low input capacitance. This propagate term governs two GDI and mGDI-implemented multiplexer-style structures [19].

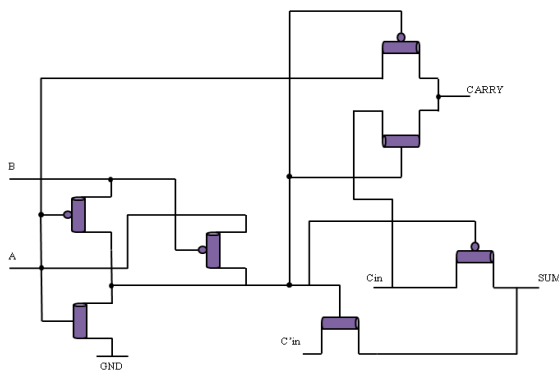


Fig. 4. GDI Based Full Adder. [19]

But With the pMOS source connected to input A and the nMOS source connected to the carry-in, the mGDI-based 2T multiplexer generates the carry output. When $X=0$ (i.e., $A = B$), the multiplexer selects A, resulting in the correct generate/kill output; when $X=1$ (i.e., $A \neq B$), the multiplexer propagates Cin , providing the correct carry result. The mGDI technique minimizes body impact and keeps the carry signal close to rail-to-rail levels by connecting wells to their respective sources. A transmission gate selector that selects between Cin and Cin bar is used to generate the sum output. Thus, the total $= A \oplus B \oplus Cin$ is correctly realized when $X=0$ and when $X=1$, respectively, the total crosses Cin and Cin bar. To compensate for threshold voltage loss across the pass devices and guarantee full-swing logic levels without causing contention, a weak level-restoring pMOS transistor is added at the sum output. Thus, only nine transistors are needed for the entire adder: two for the carry multiplexer, two for the sum selector, one for the restorer, and four for the XOR/XNOR core. By examining every input instance, it is possible to confirm that the design is accurate and that both sum and carry match the canonical full-adder truth table. This 9T full adder maintains high speed and smooth output transitions while achieving lower area and power than traditional designs. It is ideal for

low-power VLSI applications, like arithmetic circuits in portable and energy-efficient systems, due to its small size, low dynamic power, and quick carry generation.

III. MODIFIED FULL ADDER

The accompanying diagram's modified 9-transistor (9T) full adder design combines Transmission Gate Logic (TGL), Gate Diffusion Input (GDI), and modified-GDI (mGDI) approaches to achieve basic adder functionality. The circuit is specifically designed to use a TGL-based XOR/XNOR network to effectively generate the propagate term $(A \oplus B)$. This method guarantees little delay in the delivery of both the signal and its complement. An mGDI-based, two-transistor multiplexer is used to generate the carry output. The conventional carry function, $Cout = A \cdot B + Cin \cdot (A \oplus B)$, is implemented in this step by using the propagate signal to dynamically choose between input A and the carry-in (Cin). $SUM = A \oplus B \oplus Cin$ is the result of selecting between Cin and its complement using a GDI-based multiplexer that is controlled by the propagate signal. This produces the proper XOR behavior then.

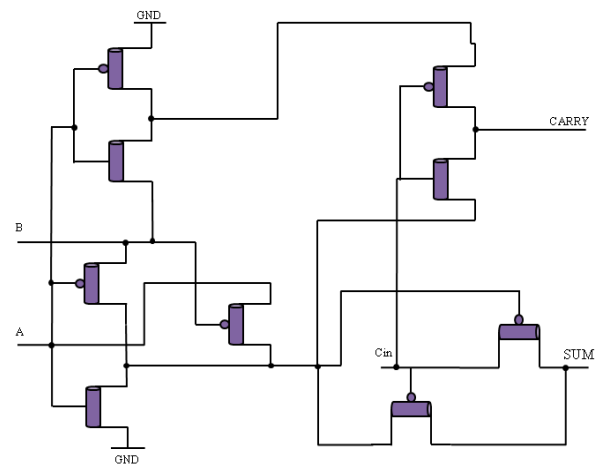


Fig. 5. Modified Circuit.

By guaranteeing complete swing at the sum node, a weak level-restoring transistor removes the threshold voltage drop problems that come with pass-transistor logic. This change ensures compactness without compromising full-adder functionality by slightly rearranging the transistor network while keeping the transistor count at 9T. By minimizing lengthy logic paths and lowering internal node capacitances, this arrangement has the advantage of achieving low power consumption and high-speed operation while preserving near rail-to-rail outputs. The modified 9T full adder is hence appropriate for low-power VLSI design applications since it offers an effective trade-off between power, performance, and area.

IV. PROPOSED FULL ADDER

The suggested circuit simplifies the implementation of the propagate function, hence altering the previous 9T full adder. Prior systems used a TGL-based XOR/XNOR block to generate the propagate term $(A \oplus B)$, and GDI and mGDI multiplexers were driven by this output for SUM and CARRY production. Additionally, a weak pMOS restorer was needed to handle threshold voltage loss

in the SUM output, increasing the circuit's complexity and power consumption.

On the other hand, the suggested architecture does not require a separate XOR block and restorer transistor because it incorporates the propagate function directly into the transistor-level GDI network. Compact GDI structures are used to generate SUM and CARRY, which result in near full-swing levels, decreased logic depth, enhanced symmetry, and a lower power–delay product. Because of this, the architecture is more effective, smaller, and ideal for low-power arithmetic applications.

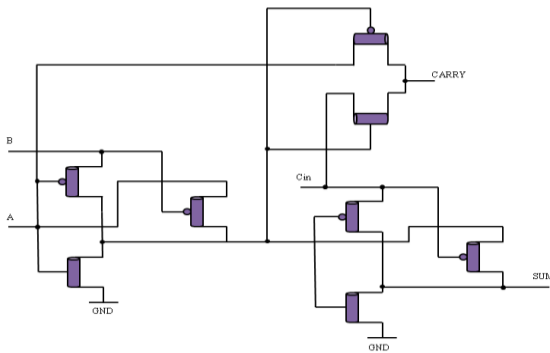


Fig. 6. Proposed Full Adder.

Let $X=A\oplus B=AB'+A'B$

Carry (Cout):

$$\text{Cout}=A\cdot B+C_{in}\cdot(A\oplus B)\rightarrow(1)$$

$$\text{Cout}=X'\cdot A+X\cdot C_{in}\rightarrow(2)$$

(when $X=0$ the cell selects A, when $X=1$ it selects C_{in}).

Sum:

$$\text{SUM}=A\oplus B\oplus C_{in}=X\oplus C_{in}\rightarrow(3)$$

TABLE 1. Functional Table of Full Adder.

A	B	Cin	SUM	CARRY
0	0	0	0	0
0	0	1	1	0
0	1	0	1	0
0	1	1	0	1
1	0	0	1	0
1	0	1	0	1
1	1	0	0	1
1	1	1	1	1

V. RESULTS AND DISCUSSION

Using HSPICE software, the current and suggested approaches are designed and simulated. HSPICE program assists in obtaining the delay, dynamic power, EDP, and PDP for the proposed and current works.

5.1 Comparing Simulation Results of all the Reference circuits

TABLE 2 compares the transistor count, average power consumption, latency, Power latency Product (PDP), and Energy Delay Product (EDP) of various full adder (FA) circuits. Because they use a comparatively large number of

transistors (26), the Hybrid Full Adder (HFA) and Binary Full Adder (BFA) have larger areas and more sophisticated designs. While the HFA exhibits reasonable power consumption (8.709×10^{-6} W) and latency (3.410×10^{-10} s), the BFA's performance is unstable due to an abnormal negative delay.

Among all reference circuits, the typical CMOS design with 28 transistors achieves the lowest PDP (3.03×10^{-17}) and EDP (9.96×10^{-28}) due to its extremely low power consumption (9.248×10^{-7} W) and delay (3.281×10^{-11} s). But in large-scale integration, its high transistor count limits its use for small VLSI designs by increasing area and power dissipation. However, the 9T adder has the lowest EDP (1.29×10^{-30}) and PDP (4.20×10^{-18}), indicating improved energy efficiency, albeit at the expense of a larger average power consumption (1.367×10^{-5} W). With just eight transistors needed for implementation, the suggested architecture strikes a fair compromise between delay and power. It consumes marginally less power (1.229×10^{-5} W) than the BFA and 9T designs while offering a much reduced delay (5.550×10^{-11}) compared to the HFA and BFA circuits. The suggested design offers higher energy-delay efficiency with little transistor consumption, as confirmed by the PDP (6.73×10^{-16}) and EDP (3.74×10^{-26}) values. This makes it ideal for low-power and area-constrained VLSI applications. The comparison graph is shown in below Fig. 7 and Fig. 8.

Fig. 7. Comparison graph between Average Power and Full Adders.

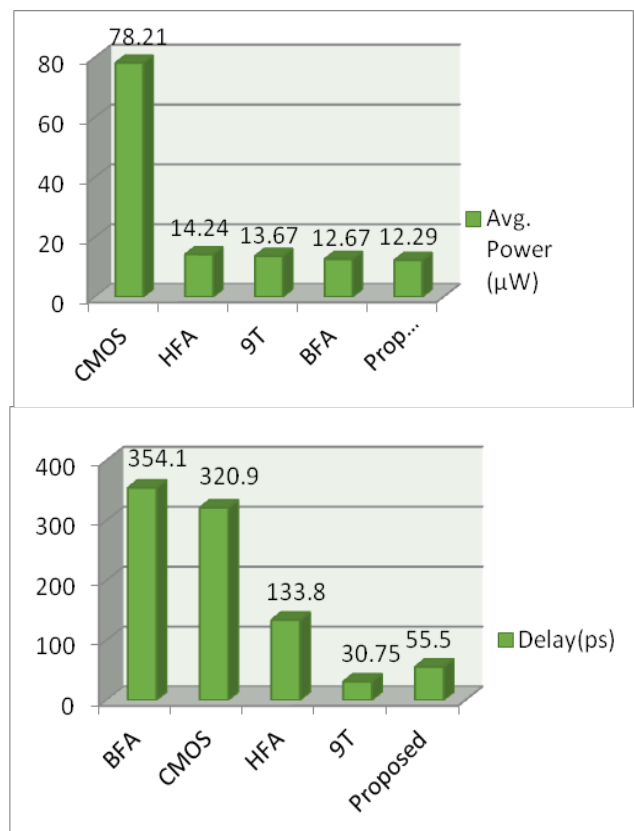


Fig. 8. Comparison graph between Delay and Full Adders.

5.2 Analysis of Voltage Scaling for the Suggested Complete Adder

As seen in TABLE 3, the performance of the suggested full adder was further assessed by changing the supply voltage from 0.7 V to 1.5 V. All voltage levels show that the delay stays at 5.550×10^{-11} s, suggesting that the circuit maintains consistent timing performance despite changes in the supply voltage. For VLSI systems to operate dependably, this stability is ideal. Higher supply voltage causes an increase in average power consumption, which ranges from 1.197×10^{-5} W at 0.7 V to 1.374×10^{-5} W at 1.5 V, as predicted. As a result of their direct reliance on power, PDP and EDP both rise with voltage. The PDP and EDP values are 6.65×10^{-16} and 3.69×10^{-26} at 0.7 V and 7.63×10^{-16} and 4.23×10^{-26} at 1.5 V, respectively.

5.3 Circuit Metric Variation with Temperature

TABLE 4 summarizes the findings of an analysis of the suggested circuit's performance at various temperatures. At lower temperatures, the average power consumption is found to be essentially constant, but at higher temperatures, it increases dramatically. At 1000°C , the maximum power of 1.677×10^{-5} W was recorded. With the lowest value of 3.425×10^{-10} , 3.425×10^{-10} s at 1000°C and the greatest value of 7.461×10^{-10} , 7.461×10^{-10} s at 50°C , the propagation delay exhibits discernible fluctuations. Similar patterns can be seen in the Power-Delay Product (PDP) and Energy-Delay Product (EDP), two crucial metrics of merit for circuit performance. The EDP drops from 6.82×10^{-24} to 1.97×10^{-24} across the same temperature range, whereas the PDP lowers from 9.14×10^{-15} at 50°C to 5.74×10^{-15} at 1000°C respectively. This makes the suggested circuit a more reliable option because it successfully maintains efficiency and enhanced energy performance even at greater operating temperatures, demonstrating its appropriateness for real-world applications.

5.4 Variation in Power Consumption with Supply Voltage

The data in the accompanying table and graph shows how supply voltage and power consumption are related. The results show that as the supply voltage goes from 0.7 V to 1.5 V, power dissipation gradually increases from 1.197×10^{-5} W to 1.374×10^{-5} W. This demonstrates unequivocally that there is a direct correlation between the applied voltage and power consumption. The plotted graph provides a visual representation of this behavior. The voltage levels are plotted by the blue line, and the equivalent power dissipation is shown by the red line. Even while the graph indicates a steady upward trend in power consumption as voltage increases, the rate of this development becomes more obvious and prominent at higher voltages, especially once the threshold of 1V is exceeded.

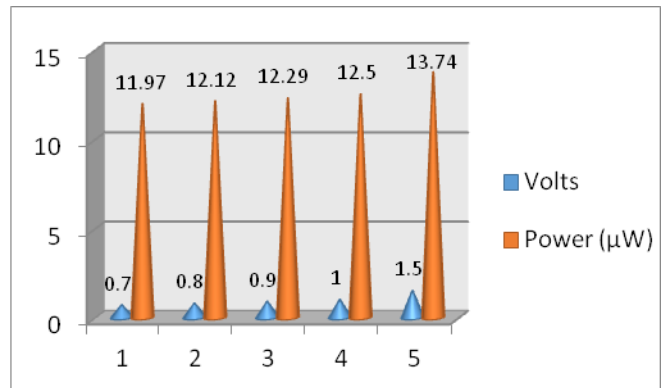
TABLE 5. Variation in Power Consumption with Supply voltage

Volts	Power (μW)
0.7	11.97
0.8	12.12
0.9	12.29
1	12.5
1.5	13.74

In circuit design, this analysis emphasizes how crucial supply voltage optimization is to striking a balance

between performance and power efficiency. The accompanying Table 5 and Figure 9 provide a visual representation of this relationship.

Fig. 9. The graph shows how voltage and power are related.



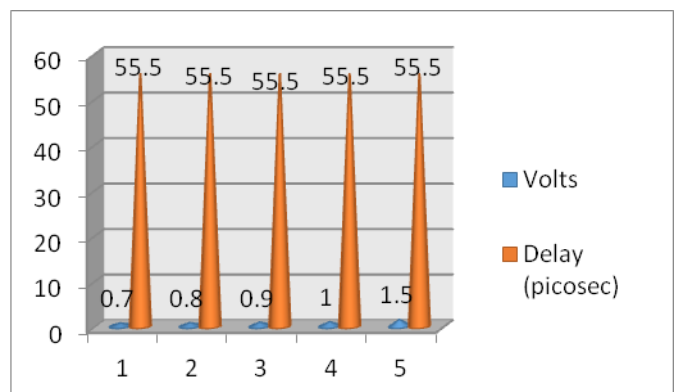
5.5 Analysis of Relation between Delays and Supply Voltage

The propagation delay stays constant at 5.55 ns over a range of supply voltages from 0.7 V to 1.5 V, according to data shown in Table 1 and Figure 1. This shows that the suggested design offers constant timing behavior and reliable delay performance regardless of operating voltage variations, which is advantageous for low-power circuit design

Volts	Delay (picosec)
0.7	55.50
0.8	55.50
0.9	55.50
1	55.50
1.5	55.50

TABLE 6. Variation in Power Consumption with Supply voltage

Fig. 10. The graph shows how Voltage and Delay are related.



The circuit's delay stays flat and steady despite rising voltage levels, as shown by a graph and table, suggesting stable timing characteristics appropriate for reliable digital applications is shown in above TABLE 6 and Fig.10.

5.7 Temperature's Impact on Average Power

The proposed 8T full adder design's average power consumption varies with different operating temperatures; Table 2 and Figure 2 of the project provide specific results. In particular, a wide temperature range—from a mild 25

degrees Celsius to a severe 1000 degrees Celsius—is taken into consideration in the research.

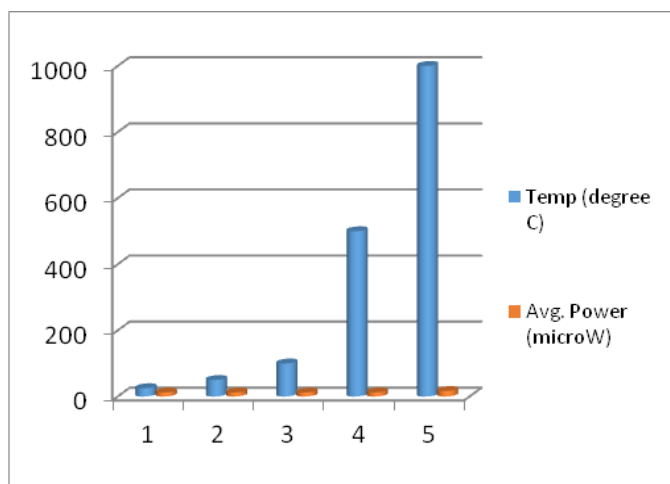
The average power consumption in the lower temperature range—that is, between 25°C and 100°C—is remarkably constant, fluctuating only slightly between 1.215 and 1.23 watts. As the temperature climbs further, a steady increase in power usage is seen, eventually reaching 1.257 Watts at 500°C. When the temperature reaches 1000°C, the average power consumption increases significantly to 1.677 Watts. This pattern demonstrates how the suggested design keeps power performance steady and effective at typical operating temperatures (up to about 100°C), which is essential for practical VLSI applications. The anticipated effect of leakage currents and device instability at extreme thermal conditions is demonstrated by the steep spike at very high temperatures.

The findings therefore verify that the suggested adder is dependable for nanoscale digital systems functioning in a range of thermal conditions since it provides strong low-power performance within realistic temperature constraints. This is shown in below TABLE 7 and Fig. 11.

TABLE 7. Temperature's Impact on Average Power

Temp (degree C)	Avg. Power (microW)
25	12.30
50	12.25
100	12.15
500	12.57
1000	16.77

Fig. 11. Graph between Temperature and Supply Voltage.



5.8 Temperature's Impact on Delay

The propagation delay variation of the suggested 8T full adder with regard to various operating temperatures,

spanning from 25°C to 1000°C, is shown in TABLE 8 and Figure 12.

The delay is shown to behave non-linearly as the temperature varies. The delay increases somewhat at lower temperatures, reaching 6.803s at 100°C and 7.461 s at 50°C. The delay, however, diminishes with increasing temperature, falling to 5.212 s at 500°C and sharply decreasing to 3.425 s at 1000°C.

Carrier mobility and threshold voltage alterations in CNTFET devices at various temperatures are responsible for this variation. Reduced mobility at moderate temperatures causes the delay to increase initially, whereas improved conduction effects in CNT channels may be the cause of the subsequent decrease at higher temperatures.

Overall, the results reveal that under realistic operating temperatures (25°C to 100°C), the suggested adder exhibits STABLE and reasonable delay performance; under high thermal circumstances, however, the delay is reduced. This implies that the design is appropriate for integrating digital circuits at the nanoscale since it can sustain sTABLE timing behavior in a variety of temperature conditions.

TABLE 8. Temperature's Impact on Delay

Temp (degree C)	Delay(pico sec)
25	5.494
50	7.461
100	6.803
500	5.212
1000	3.425

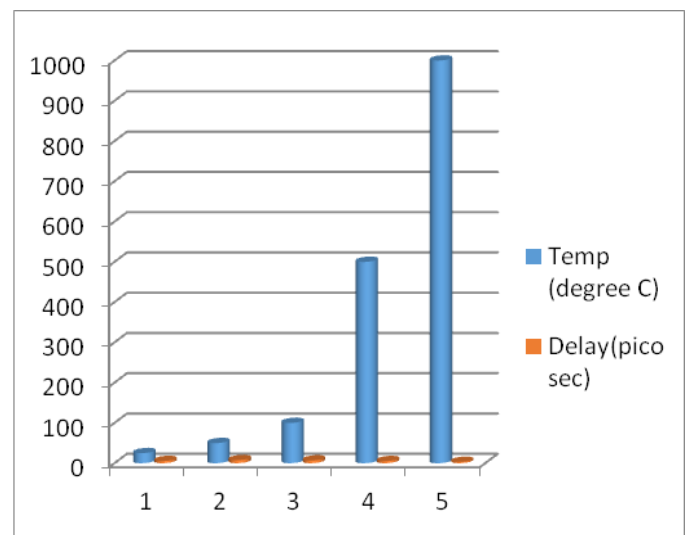


Fig. 12. Graph between Temperature and Delay.

TABLE 2. Comparing simulation of all the reference circuits

FA	Tran. Count	Avg. Power(μ W)	Delay(ps)	PDP	EDP
HFA	26	78.21×10^{-6}	133.8×10^{-12}	1.04×10^{-14}	1.40×10^{-21}
CMOS	28	14.24×10^{-6}	320.9×10^{-12}	4.56×10^{-15}	1.46×10^{-21}
BFA	26	12.67×10^{-6}	354.1×10^{-12}	4.48×10^{-15}	1.58×10^{-21}
9T	9	13.67×10^{-6}	30.75×10^{-12}	4.20×10^{-16}	1.29×10^{-22}
Proposed	8	12.29×10^{-6}	55.50×10^{-12}	6.82×10^{-16}	3.78×10^{-23}

TABLE 3. Analysis scaling for the suggested complete adder

Volts	Avg. Power (μ W)	Delay (ps)	PDP	EDP
0.7	11.97×10^{-6}	55.50×10^{-12}	6.65×10^{-16}	3.69×10^{-26}
0.8	12.12×10^{-6}	55.50×10^{-12}	6.73×10^{-16}	3.74×10^{-26}
0.9	12.29×10^{-6}	55.50×10^{-12}	6.82×10^{-16}	3.79×10^{-26}
1.0	12.50×10^{-6}	55.50×10^{-12}	6.94×10^{-16}	3.85×10^{-26}
1.5	13.74×10^{-6}	55.50×10^{-12}	7.63×10^{-16}	4.23×10^{-26}

TABLE 4. Circuit Metric Variation with Temperature Supply Voltage

Temp (degree C)	Avg. Power (μ W)	Delay (ps)	PDP	EDP
25	12.30×10^{-6}	549.4×10^{-12}	6.76×10^{-15}	3.71×10^{-24}
50	12.25×10^{-6}	746.1×10^{-12}	9.14×10^{-15}	6.82×10^{-24}
100	12.15×10^{-6}	680.3×10^{-12}	8.27×10^{-15}	5.63×10^{-24}
500	12.57×10^{-6}	521.2×10^{-12}	6.55×10^{-15}	3.42×10^{-24}
1000	16.77×10^{-6}	342.5×10^{-12}	5.74×10^{-15}	1.97×10^{-24}

VI. CONCLUSION

In this work, a novel 8-transistor complete adder design that targets low power consumption, decreased size, and efficient delay performance was realized in 32 nm CNTFET technology. Because of its dependable performance under voltage scaling, the suggested 8T full adder architecture exhibits a consistent propagation delay of 5.55×10^{-11} s throughout a supply voltage range of 0.7 V to 1.5 V, making it appropriate for low-power applications. It offers a balance between transistor count, power consumption, and energy efficiency when compared to other adder architectures, demonstrating enhanced Power-Delay Product (PDP) and Energy-Delay Product (EDP) for reduced silicon area and increased performance. A better balance between transistor count, power, delay, and energy efficiency is offered by the suggested design. With an 8-transistor implementation that provides a full-swing output, consistent latency, good thermal stability, and durability under manufacturing variances, the architecture beats prior solutions (CMOS, hybrid, GDI). The 8T CNTFET full adder is positioned as the best option for next low-power VLSI systems thanks to these qualities.

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